Response to Reviewers

Camera-Motion Compensation for DeepSORT Person Tracking: A Controlled Study of Benefit and Cost Across Camera Motions

Omar Hussain and Zayed Mohammed

We thank the reviewers for their careful reading. The comments led us to new experiments that test the original controlled study from several sides and show how the benefit of camera-motion compensation (CMC) depends on the type of camera motion, including the PTZ and pole-mounted cameras used on sites. The title was extended accordingly. New work: virtual PTZ sweeps and pole sway over seven static scenes with two detectors; a real sweeping camera (13 SportsMOT clips); compensation from PTZ angle readings; a simulated PPE-attribution analysis; a time-aware Kalman check for frame skipping; a leave-one-out analysis; a repeat-run check; and pre-registered tests on three independent MOT15 moving sequences with two detectors. Protocols were recorded before each new set of runs. Two earlier claims were weakened (the frame-skipping gain and generalisation to forward-moving cameras) and are now reported as boundary cases. The revised manuscript is 6 pages including references; numbers below refer to it (S-tables are in the supplementary material).

#	Comment	Response and change
M1	Only four moving sequences	We added a real sweeping camera (13 SportsMOT clips: +1.43 HOTA, interval [0.55, 1.81], Table II), seven scenes for the virtual camera (Table I), three independent MOT15 moving sequences (Table IV; four MOT15 sequences were excluded as image-identical to MOT17) and a leave-one-out analysis of MOT17 (+0.28 to +0.57 with any sequence removed, Sec. V-A). On the independent, mostly forward-moving sequences CMC did not help (IDF1 −1.61 with YOLOv8n; not significant with ACF detections); we report this as a boundary case.
M2	Frame-level bootstrap	We state that intervals describe uncertainty within the observed sequences, explain why scene-level resampling is not informative with few sequences, and add the leave-one-out analysis, the sign test and the per-clip SportsMOT results (Table II, Table S6) as between-sequence evidence (Sec. IV).
M3 / m1	Kalman Δt when subsampling	We document the per-step convention (Sec. III, "Time step") and ran a time-aware model ($\Delta t = k$, process noise $\times k$, max. age $\lceil 30/k \rceil$) with both detectors and bootstrap intervals (Table S11). The baseline improved by 5–9 HOTA at $k = 6$ and the CMC gain shrank (ECC +0.69 and +2.74 instead of +5.43 and +10.12). Varying only the maximum age left the gains intact, so the effect comes from Δt and noise scaling. Abstract, discussion and conclusion now say that frame-skipping gains depend on the Kalman time step.
M4 / m9	Hybrid tuned on test data	The hybrid is labelled exploratory throughout (Algorithm 1, captions, Table S10). On the independent sequences it was not better than affine (Table IV); on SportsMOT it gave the largest mean gain but no interval was computed. We make no generalisation claim.
M5 / m4	Cross-machine differences	Repeating CMC estimation and tracking on one machine from the same cached detections gave bit-identical outputs for six synthetic variants. The divergence therefore arises at or before detection (different CPU architecture, Python and ONNX Runtime); the first environment no longer exists, so the exact stage is not confirmed (Sec. V-G). Software versions are listed in Sec. IV.
M6 / m2 / m3	Missing method details	Added a parameter table (Table S7) and hybrid pseudocode with gates, ECC fallback and identity fallback (Algorithm 1). The estimator is described as a four-parameter similarity transform; the virtual camera is defined in Sec. IV.

#	Comment	Response and change
M7	No modern-tracker comparison	Not added: ByteTrack, BoT-SORT and StrongSORT change association as well as motion, which would confound the controlled question (Sec. VI). Instead we extended the study towards deployment, giving a practical rule for when CMC is worth enabling (Sec. VI).
M8	Ethics statement	Added a dedicated Ethics Statement based on the public-data study. We do not claim institutional approval or formal exemption.
M9 / m12	PPE claims untested	Added a simulated PPE-attribution analysis of Eq. 1 (Table III): CMC roughly halved duplicate violation reports at a $40°/s$ sweep (interval of the change $[-2.35, -0.22]$) but did not reduce misattribution. It is labelled simulated. We also added a planned evaluation with consented PTZ site footage and cite the CMOT construction dataset (mostly fixed cameras).
m5	Timing hardware	Table S9 caption and footnote state the machine, run and timing method; the 7% overhead is labelled an estimate from component medians.
m6	Oracle details	Sec. IV gives the oracle transform for synthetic crops and for the virtual camera (least-squares similarity fit to the true homography), visibility scaling and edge handling, and confirms that all conditions share cached detections.
m7	Null hypothesis, corrections	Sign-test null stated; no multiple-comparison correction was applied, and secondary intervals are described as descriptive (Sec. IV).
m8	Per-sequence intervals	Per-sequence MOT17 intervals are in Table S4; intervals were added for the time-aware, independent, surveillance and SportsMOT experiments; per-clip SportsMOT results are in Table S6.
m10	More qualitative examples	Added a real moving-camera example from MOT17-13 (Fig. 1); the synthetic example moved to the supplementary material.
m11	Code link and licence	Added a Data and Code Availability section; the archive contains code, models, protocols, cached inputs, outputs and bootstrap samples. A permanent link will be added when the repository is published.
m13	Typos and captions	Captions and abbreviations checked; the evidence table (Table S10) separates measured-once (S), inconclusive (I) and untested (N) results.